Frequency $n = \frac{\omega}{2\pi} = \frac{1}{2\pi}$ hertz

Time period $T = \frac{1}{n} = 2\pi$ seconds.

Example 1.4. *The motion of a particle in simple harmonic motion is given by $x = a \sin \omega t$. If it has a speed u when the displacement is x_1 and speed v when the displacement is x_2, show that the amplitude of the motion is*

$$a = \left[\frac{v^2 x_1^2 - u^2 x_2^2}{v^2 - u^2}\right]^{\frac{1}{2}}$$

(Utkal, 1989)

Here $$x = a \sin \omega t$$

$$u = \frac{d x_1}{dt} = \omega \sqrt{a^2 - x_1^2} \quad \ldots (i)$$

and $$v = \frac{d x_2}{dt} = \omega \sqrt{a^2 - x_2^2} \quad \ldots (ii)$$

Squaring and dividing

$$\frac{u^2}{v^2} = \frac{a^2 - x_1^2}{a^2 - x_2^2}$$

$$u^2 a^2 - u^2 x_2^2 = v^2 a^2 - v^2 x_1^2$$

$$a^2 [v^2 - u^2] = v^2 x_1^2 - u^2 x_2^2$$

$$a = \left[\frac{v^2 x_1^2 - u^2 x_2^2}{v^2 - u^2}\right]^{\frac{1}{2}} \quad \ldots (iii)$$

Example 1.5. *Show that for a particle executing SHM, the instantaneous velocity is $\omega \sqrt{a^2 - y^2}$ and instantaneous acceleration is $-\omega^2 y$.*

For a particle executing SHM,

$$y = a \sin (\omega t + \alpha) \quad \ldots (1)$$

The instantaneous velocity,

$$v = \frac{dy}{dt} = + a \omega \cos (\omega t + \alpha) \quad \ldots (2)$$

From equation (1)

$$\sin (\omega t + \alpha) = \frac{y}{a}$$

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Here
$$n_1 = 2{\cdot}7 \times 10^{19}$$
$$P_1 = 76 \times 13{\cdot}6 \times 980 \text{ dynes/cm}^2$$
$$P_2 = 10^{-6} \text{ mm of Hg}$$
$$= 10^{-7} \text{ cm of Hg}$$
$$P_2 = 10^{-7} \times 13{\cdot}6 \times 980 \text{ dynes/cm}^2$$
$$n_2 = \frac{2{\cdot}7 \times 10^{19} \times 10^{-7} \times 13{\cdot}6 \times 980}{76 \times 13{\cdot}6 \times 980}$$
$$n_2 = \mathbf{3{\cdot}553 \times 10^{10}}$$

(*ii*) Let the R.M.S. velocity at 0°C be C_2 and at 39°C be C_3. Number of molecules per cc at 0°C and 10^{-6} mm of Hg pressure

$$= n_2$$

and at 39°C and 10^{-6} mm of Hg pressure

$$= n_3$$

$$P = \frac{1}{3} mnC^2$$

Here pressure is the same in both the cases

$$\therefore \quad \frac{1}{3} mn_2C_2^2 = \frac{1}{3} mn_3C_3^2$$

$$\therefore \quad n_2C_2^2 = n_3C_3^2$$

$$n_3 = n_2 \frac{C_2^2}{C_3^2}$$

But
$$C \propto \sqrt{T}$$

$$\therefore \quad \frac{C_2^2}{C_3^2} = \frac{T_2}{T_3}$$

$$\therefore \quad n_3 = \frac{n_2 \times T_2}{T_3}$$

Here
$$n_2 = 3{\cdot}553 \times 10^{10}$$
$$T_2 = 0° \text{ C}$$
$$= 273 \text{ K}$$
$$T_3 = 39° \text{ C}$$
$$= 273 + 39$$
$$= 312 \text{ K}$$

$$\therefore \quad n_3 = \frac{3{\cdot}553 \times 10^{10} \times 273}{312}$$

$$n_3 = \mathbf{3{\cdot}109 \times 10^{10}}$$

5·19 Avogadro's Hypothesis

Consider two gases A and B at a pressure P and each having a volume V.

Mass of each molecule of the first gas $= m_1$

Number of molecules of the first gas $= n_1$

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$$\therefore \quad C_v.P.dV + C_v.V.dP + C_p.PdV - C_vPdV = 0$$

$$C_p.P.dV + C_v.V.dP = 0$$

Dividing by $C_v.PV$,

$$\frac{C_p}{C_v} \cdot \frac{dV}{V} + \frac{dP}{P} = 0$$

But $$\frac{C_p}{C_v} = \gamma$$

$$\therefore \quad \frac{dP}{P} + \gamma \frac{dV}{V} = 0$$

Integrating, $\log P + \gamma \log V = \text{const.}$

$$\log PV^\gamma = \text{const.}$$

or $$PV^\gamma = \text{const.} \quad \text{...(iv)}$$

This is the equation connecting pressure and volume during an adiabatic process.

Taking $$PV = rT$$

or $$P = \frac{rT}{V}$$

$$\left(\frac{rT}{V}\right) . V^\gamma = \text{const.}$$

But r is const.

$$rTV^{\gamma-1} = \text{const.}$$

$$\therefore \quad TV^{\gamma-1} = \text{const.} \quad \text{...(v)}$$

Also $$V = \frac{rT}{P}$$

$$P\left[\frac{rT}{P}\right]^\gamma = \text{const.}$$

or $$\frac{r^\gamma T^\gamma}{P^{\gamma-1}} = \text{const.}$$

or $$\frac{P^{\gamma-1}}{T^\gamma} = \text{const.} \quad \text{...(vi)}$$

Thus, during an adiabatic process

(i) $PV^\gamma = \text{const.}$

(ii) $TV^{\gamma-1} = \text{const.}$ and

(iii) $\dfrac{P^{\gamma-1}}{T^\gamma} = \text{const.}$

Example 6·2. *A motor car tyre has a pressure of 2 atmospheres at the room temperature of 27°C. If the tyre suddenly bursts, find the resulting temperature.*

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Here,
$$P_1 = 2 \text{ atmospheres}$$
$$T_1 = 273+27$$
$$= 300 \text{ K}$$
$$P_2 = 1 \text{ atmosphere}$$
$$T_2 = ?$$
$$\gamma = 1{\cdot}4$$
$$\frac{P_1^{\gamma-1}}{T_1^{\gamma}} = \frac{P_2^{\gamma-1}}{T_2^{\gamma}}$$
$$\left(\frac{P_2}{P_1}\right)^{\gamma-1} = \left(\frac{T_2}{T_1}\right)^{\gamma}$$
$$\left(\frac{1}{2}\right)^{0.4} = \left(\frac{T_2}{300}\right)^{1.4}$$
$$0{\cdot}4 \log (0.5) = 1{\cdot}4 [\log T_2 - \log 300]$$
$$-0{\cdot}1204 = 1{\cdot}4 \log T_2 - 3{\cdot}4680$$
$$1{\cdot}4 \log T_2 = 3{\cdot}4680 - 0{\cdot}1204$$
$$= 3{\cdot}3476$$
$$\log T_2 = \frac{3{\cdot}3476}{1{\cdot}4}$$
$$= 2{\cdot}3911$$
$$T_2 = 246{\cdot}1 \text{ K}$$
$$= -26{\cdot}9°\text{C}$$

Example 6·3. *A quantity of air at 27°C and atmospheric pressure is suddenly compressed to half its original volume. Find the final (i) pressure and (ii) temperature.*

(i) $P_1 = 1$ atmosphere ; $P_2 = ?$, $\gamma = 1{\cdot}4$

$V_1 = V$; $V_2 = \frac{V}{2}$

During sudden compression, the process is adiabatic
$$P_1V_1^{\gamma} = P_2V_2^{\gamma}$$
$$P_2 = P_1\left[\frac{V_1}{V_2}\right]^{\gamma}$$
$$= 1[2]^{1.4}$$
$$= 2{\cdot}636 \text{ atmospheres}$$

(ii) $V_1 = V$; $V_2 = \frac{V}{2}$
$$T_1 = 300 \text{ K} ; T_2 = ?$$
$$\gamma = 1{\cdot}4$$
$$T_1(V_1)^{\gamma-1} = T_2 (V_2)^{\gamma-1}$$
$$T_2 = T_1[2]^{1.4-1}$$
$$= 300[2]^{0.4}$$
$$= 395{\cdot}9 \text{ K}$$
$$= 122{\cdot}9°\text{C}$$

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$$= \frac{\delta H}{T} = ms \int_{T_1}^{T} \frac{dT}{T}$$

$$= 10 \times 1 \times \log_e \frac{303}{333}$$

$$= 10 \times 2{\cdot}3026 \times \log_{10} \left(\frac{303}{333}\right)$$

$$= -0{\cdot}9443 \text{ cal/K}^\circ$$

Nagative sign indicates decrease in entropy.

(*iii*) Change in entropy when the temperature of 30 g of water change from 293 K° to 303 K°,

$$= \frac{\delta H}{T} = ms \int_{T_2}^{T} \frac{dT}{T}$$

$$= 30 \times 1 \cdot \log_e \frac{303}{293}$$

$$= 30 \times 2{\cdot}3026 \log_{10} \frac{303}{293}$$

$$= +1{\cdot}002 \text{ cal/K}^\circ$$

The total gain in entropy of the system

$$= +1{\cdot}0020 - 0{\cdot}9443$$

$$= \mathbf{0{\cdot}0577 \text{ cal/K}^\circ}$$

Example A-17 *Calculate the increase in entropy of 10 kg of water at 100°C when it changes to vapour.* [*Delhi (Hons) 1983*]

Increase in entropy when 10 kg of water at 100°C is changed into vapour

$$dS = \frac{\delta H}{T} = \frac{10 \times 10^3 \times 1 \times 540}{373}$$

$$dS = \mathbf{14{,}477 \text{ cal/K}}$$

Example A-18 *When 50 g of water is heated from 10°C to 90°C, by how much does its entropy change ?* [*Delhi, 1983*]

Here $T_1 = 273 + 90 = 363$ K

$T_2 = 273 + 10 = 283$ K

Increase in entropy when 50 g of water is heated from 10°C to 90°C

$$dS = \int_{T_1}^{T_2} \frac{\delta H}{T}$$

$$dS = ms \times 2{\cdot}3026 \log \left(\frac{T_2}{T_1}\right)$$

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$$dS = 50 \times 1 \times 2{\cdot}3026 \log \left(\frac{363}{283}\right)$$

$$dS = \mathbf{12{\cdot}448\ cal/K}$$

Example A-19 *Calculate the change in entropy when 10 grams of ice at 0°C is converted into steam at 100°C.* [*Delhi, 1981*]

(1) Increase in entropy when 10 grams of ice at 0°C is converted into water at 0°C.

$$dS = \frac{\delta H}{T}$$

$$= \frac{10 \times 80}{273} = 2{\cdot}93 \text{ cal/K}$$

(2) Increase in entropy when the temperature of 10 g of water is raised from 0°C to 100°C

$$dS = \int_{T_1}^{T_2} \frac{\delta H}{T}$$

$$= ms \times 2{\cdot}3026 \log_{10} \left(\frac{T_2}{T_1}\right)$$

$$= 10 \times 1 \times 2{\cdot}3026 \log_{10} \left(\frac{373}{273}\right)$$

$$= 3{\cdot}12 \text{ cal/K}$$

(3) Increase in entropy when 10 g of water at 100°C is converted into steam at 100°C

$$dS = \frac{\delta H}{T}$$

$$= \frac{10 \times 540}{373} = 14{\cdot}47 \text{ cal/K}$$

Total increase in entropy $= 2{\cdot}93 + 3{\cdot}12 + 14{\cdot}47$

$= \mathbf{20{\cdot}52\ cal/K.}$

Example A-20 *Calculate the change in entropy of 1 gram of nitrogen when its temperature rises from 50°C to 100°C while its volume is kept constant. Molar specific heat, $C_v = 0{\cdot}18$ and molecular weight of nitrogen is 28.*

[*Delhi, 1981*]

$$\text{Specific heat of nitrogen} = \frac{C_v}{28} = \left(\frac{0{\cdot}18}{28}\right)$$

$$\text{Change in entropy, } dS = \int_{T_1}^{T_2} \frac{\delta H}{T}$$

$$dS = \frac{1 \times 0{\cdot}18}{28} \int_{T_1}^{T_2} \frac{dT}{T}$$

$$dS = \left(\frac{0{\cdot}18}{28}\right) \int_{323}^{373} \frac{dT}{T}$$

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$$dS = \left(\frac{0.18}{28}\right) \times 2.3026 \log_{10}\left(\frac{373}{323}\right)$$

$$dS = \frac{0.18 \times 2.3026 \times 0.0625}{28}$$

$$dS = 9.253 \times 10^{-4} \text{ cal/K.}$$

Example A-21 *Calculate the temperature at which ice will freeze if the pressure is increased by 135·2 atmospheres. Given that the increase in the specific volume when 1 gram of water freezes into ice is 0·091 cm³. Latent heat of fusion of ice is 80 cal/gram.*

[*Delhi, 1981*]

$$\frac{dP}{dT} = \frac{L}{T(V_2 - V_1)}$$

Here $dP = 135.2$ atmospheres

$dP = 135.2 \times 1.013 \times 10^6$ dynes/cm²

$V_2 - V_1 = -0.091$ cm³

$L = 80$ cal/g

$L = 80 \times 4.18 \times 10^7$ ergs/g

$T = 273$ K

$$\therefore \quad d = T\frac{(dP)\,T\,(V_2 - V_1)}{L}$$

$$dT = \frac{135.2 \times 1.013 \times 10^6 \times 273 \times (-0.091)}{80 \times 4.18 \times 10^7}$$

$$dT = -1 \text{ K}$$

or $dT = -1°\text{C}$

Therefore the temperature at which ice will freeze is

$-1°$C or 272 K.

Example A-22 *1 kg of water at 273 K is brought in contact with a heat reservoir at 373 K. What is the change in entropy of water as its temperature reaches 373 K ?*

[*IAS, 1983*]

Increase in entropy when the temperature of 1 kg of water is raised from 273 K to 373 K

$$dS = \int_{T_1}^{T_2} \frac{\delta H}{T}$$

$$m = 1 \text{ kg}, s = 10^3 \text{ cal/kg-K},$$

$$dS = ms \times 2.3026 \log_{10}\left(\frac{T_2}{T_1}\right)$$

$$dS = 1 \times 10^3 \times 2.3026 \log_{10}\left(\frac{373}{273}\right)$$

$$dS = 312 \text{ cal/K.}$$

Example A-23. *Calculate the change in entropy, when 0·0273 kg of ice of 0°C is converted into water at the same temperature,*

[*Delhi, 1984*]

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$$\eta_2 = 1 - \left(\frac{H_3}{H_2}\right)$$

$$\eta_1 = \eta_2$$

$$1 - \left(\frac{H_2}{H_1}\right) = 1 - \left(\frac{H_3}{H_2}\right)$$

$$1 - \left(\frac{T}{1200}\right) = 1 - \left(\frac{300}{T}\right)$$

$$\frac{T}{1200} = \frac{300}{T}$$

$$T = 600 \text{ K}$$

Example A-53 *A Carnot engine working between a source at temperature 27°C and a sink at −73°C delivers 300 calories of heat to the latter in one cycle. Calculate the work performed in joules by the engine per cycle.*

[*Calcutta, 1992*]

Here, $T_1 = 27 + 273 = 300\,\text{K}$

$T_2 = -73 + 273 = 200\,\text{K}$

$H_2 = 300$ calories.

$H_1 = ?$

$$\frac{H_1}{H_2} = \frac{T_1}{T_2}$$

$$\frac{H_1}{300} = \frac{300}{200}$$

$H_1 = 450$ calories

Work performed per cycle,

$$W = H_1 - H_2$$

$$= 450 - 300$$

$$= 150 \text{ calories}$$

$$= 150 \times 4.2 \text{ J}$$

$$= 630 \text{ J}$$

Example A-54 *In a cold storage ice melts at the rate of 3 kg. per hour when the external temperature is 28°C. Find the minimum power output of the motor used to run the refrigerator just to prevent melting of ice.*

Specific latent heat of ice fusion $= 3.3 \times 10^5 \text{ J kg}^{-1}$

[*Calicut, 1992*]

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$$m = 3 \text{ kg}$$

$$L = 3.3 \times 10^5 \text{ J kg}^{-1}$$

$$s = 4.2 \times 10^3 = \text{J kg}^{-1} - {}^\circ\text{C}$$

Heat taken by ice to melt into water and reach a temperature of28°C in one hour.

$$Q = mL + ms\theta$$

$$= 3 \times 3.33 \times 10^5 + 3 \times 4.2 \times 10^3 \times 28$$

$$= 1.343 \times 10^6 \text{ J}$$

To prevent melting of ice, the refrigerator should have the power out,

$$P = 1.343 \times 10^6 \text{ J/hour}$$

$$= \frac{1.343 \times 10^6}{3600} \text{ J/s}$$

$$= 373 \text{ Watts.}$$

Example A-55 *Calculate the depression of melting point of ice by one atmosphere increase of pressure, given, latent heat of ice* $= 3.35 \times 10^5$ J/kg *and the specific volumes of 1 kg. of ice and water at* 0°C *are* 1.090×10^{-3} m^3 *and* 10^{-3} m^3 *respectively.*

[*Kerala, 1992*]

Here,

$$L = 3.35 \times 10^5 \text{ J/kg}$$

$$T = 273 \text{ K}$$

$$V_1 = 1.090 \times 10^{-3} \text{ m}^3$$

$$V_2 = 10^{-3} \text{ m}^3$$

$$dP = 1 \text{ atmosphere}$$

$$= 0.76 \times 13.6 \times 10^3 \times 9.8$$

$$= 1.01 \times 10^5 \text{ N/m}^2$$

$$\frac{dP}{dT} = \frac{L}{T\,[V_2 - V_1]}$$

$$dT = \frac{dP \times T\,[V_2 - V_1]}{L}$$

$$= \frac{1.01 \times 10^5 \times 273\,[10^{-3} - 1.09 \times 10^{-3}]}{3.35 \times 10^5}$$

$$= -7.4 \times 10^{-3} \text{ K}$$

$$= -7.4 \times 10^{-3}\ {}^\circ\text{C}$$

Therefore, decrease in melting point of ice with an increase in pressure of one atmosphere

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$$= 7.4 \times 10^{-3}\ \text{K} = 7.4 \times 10^{-3}\ ^\circ\text{C}$$

Example A-56 *Find the change in boiling point of water for* 100 cm *change in mercury pressure if the specific volume of steam is* 1601 cm^3/gram *at* 100°C *and* 76 cm *mercury pressure. Take latent heat as* 536 cal/gram.

[*Allahabad, 1992*]

[In SI system],

Here $dP = 100\ \text{cm Hg} = 1\ \text{m Hg}$

$$= 1 \times 13.6 \times 10^3 \times 9.8\ \text{N/m}^2$$

$$= 1.33 \times 10^5\ \text{N/m}^2$$

$$V_1 = 1\ \text{cm}^3/\text{gram} = 10^{-3}\ \text{m}^3/\text{kg}$$

$$V_2 = 1601\ \text{cm}^3/\text{gram} = 1601 \times 10^{-3}\ \text{m}^3/\text{kg}$$

$$L = 536\ \text{cal/gram}$$

$$= 536 \times 4.2 \times 10^3\ \text{J/kg}$$

$$= 2.25 \times 10^6\ \text{J/kg}$$

$$T = 100^\circ\text{C} = 373\ \text{K}$$

$$dT = ?$$

$$\frac{dP}{dT} = \frac{L}{T\,[V_2 - V_1]}$$

$$dT = \frac{dP \times T\,[V_2 - V_1]}{L}$$

$$= \frac{1.33 \times 10^5 \times 373\,[1601 \times 10^{-3} - 10^{-3}]}{2.25 \times 10^6}$$

$$= \mathbf{35.28\ K}$$

$$= \mathbf{35.28^\circ C}$$

Therefore, increase in boiling point of water with an increase of 100 cm in mercury pressure

$$= \mathbf{35.28\ K = 35.28^\circ C}$$

Example A-57 *The ratio of the densities of ice and water at* 0°C *is* $\frac{10}{11}$. *Calculate the decrease in the melting point of ice, given that latent heat of fusion of ice = 80 cal/gm and change in specific volume* −0.1 c.c./gm. *The increase in pressure is one atmosphere.*

[*Osmania, 1992*]

[In SI system],

$$dP = 0.76 \times 13.6 \times 10^3 \times 9.8 = 1.01 \times 10^5\ \text{N/m}^2$$

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